WebAgent-R1: Training Web Agents via End-to-End Multi-Turn RL

Zhepei Wei, Wenlin Yao, Yao Liu, Weizhi Zhang, Qin Lu, Liang Qiu, et al.

Paper version: arXiv 2505.16421v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Training effective web agents for multi-turn interactions remains challenging due to the complexity of long-horizon decision-making across dynamic web interfaces.*

Prior web-agent RL is off-policy or iterative — replay buffers, reward models, filtering — breaking the end-to-end loop.

Motivation: three costs of off-policy RL

1. **Stale data.** Off-policy data misses state changes.
2. **Context blow-up.** Huge HTML every step.
3. **Extra machinery.** Buffers, reward models, filters.

Evaluation setup

WebArena-Lite · 165 eval · 647 RL train tasks · 9,460 BC trajectories · Qwen2.5-3B / Llama-3.1-8B · online eval · WebVoyager OOD

- **All online** SR from rule-based rubrics.
- **OOD:** WebVoyager, 25 tasks × 5 domains (3B); no seeds.

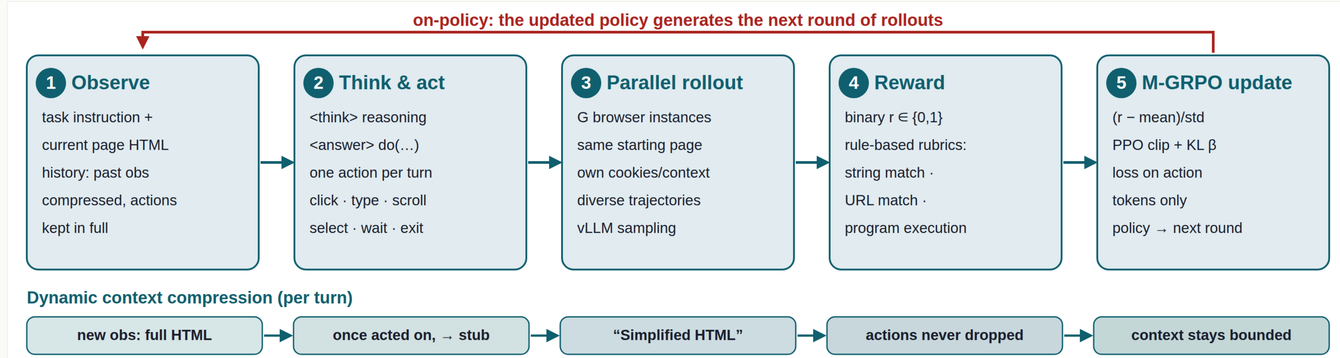
Our read (CUA Radar)

61	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; WebVoyager OOD; RL baselines; ablation.
- **Flags:** no variance; RL on same sites as eval; code unverified.
- Pattern K20. Authors' numbers, not rerun.

Method: BC warm-up, then on-policy multi-turn GRPO (M-GRPO)

- **Sees / acts:** text-only HTML of the page + task; one action per turn — do(action, argument, element) or exit(message), inside / tags.
- **Two mechanisms:** dynamic context compression (old observations → "Simplified HTML" stubs, actions kept) and parallel rollout over G browser instances per task.
- **Signal:** binary rule-based reward only; group-relative advantage across the G trajectories; loss masked to action tokens.



$$A_{i,j} = \frac{r_i - \mathrm{mean}(\mathbf{r})}{\mathrm{std}(\mathbf{r})}, \quad r_i \in \{0,1\}$$

group-relative advantage over G parallel trajectories of one task

Key results: WebArena-Lite online SR (%)

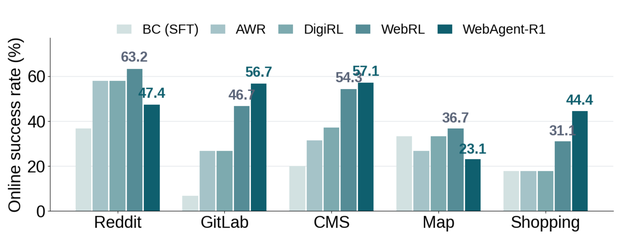


Table 2 redrawn: Llama-3.1-8B rows.

44.8 avg (8B) vs WebRL 42.4, o3 39.4

SO WHAT → *Reddit 47.4 vs 63.2; Map 23.1 vs 36.7 (WebRL).*

Ablation: RL initialization (Qwen2.5-3B)

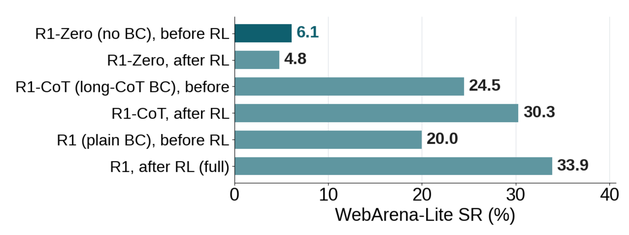


Figure 4, values from §3.4 text: before → after RL (Qwen2.5-3B).

SO WHAT → *Without BC, RL fails (6.1 → 4.8): ill-formed actions rarely earn reward. Long-CoT BC starts higher (24.5) but gains less from RL; plain BC + M-GRPO is best at 33.9.*

Headline Numbers

44.8%
WebArena-Lite SR (8B)
from 8.5 prompting · WebRL 42.4 · o3 prompting 39.4

- **33.9%** Qwen2.5-3B after RL
- **32%** WebVoyager OOD avg
- **39.4%** OpenAI o3 prompting

The paper's own controls

CONTROL	RESULT	CONTEXT
BC init (3B)	20.0 → 33.9	no BC: 6.1 → 4.8
Long-CoT BC	24.5 → 30.3	smaller RL gain
Thinking fmt	3.2 → 6.1	o4-mini 15.9 → 36.9
vs WebRL	ckpt init	same 647 tasks
Eval	165 tasks	OOD avg 32%

Takeaway

BC warm-up + multi-turn GRPO on binary rule rewards: Llama-3.1-8B reaches 44.8 on WebArena-Lite (WebRL 42.4, o3 39.4), plus OOD gains on WebVoyager — no reward model, no replay buffer.

compressed context · G browsers per task · no reward model · text-only HTML · 8×A100, full FT · more turns → higher SR

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Paper · Code